

HikarIa : Evaluation of Visual Large Language Models for photograph description

TEKLI

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MUSÉE
GUIMET

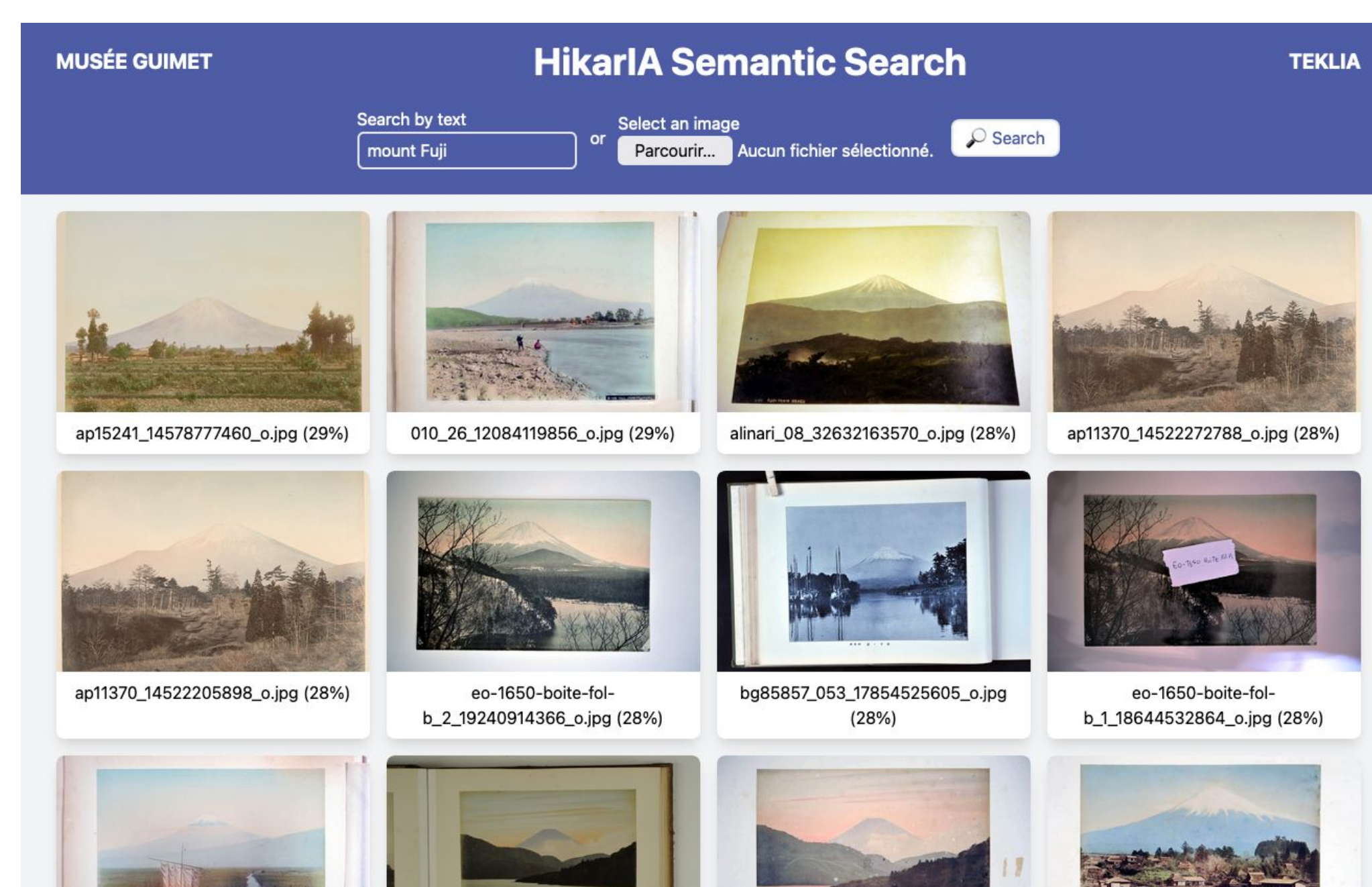
GENERAL CONTEXT

DATA

- 20,000 photographs of Japan from the 19th and early 20th centuries
- Including 19,000 collected by Joseph Dubois and acquired by the Musée Guimet in 2007

GENERAL PURPOSE

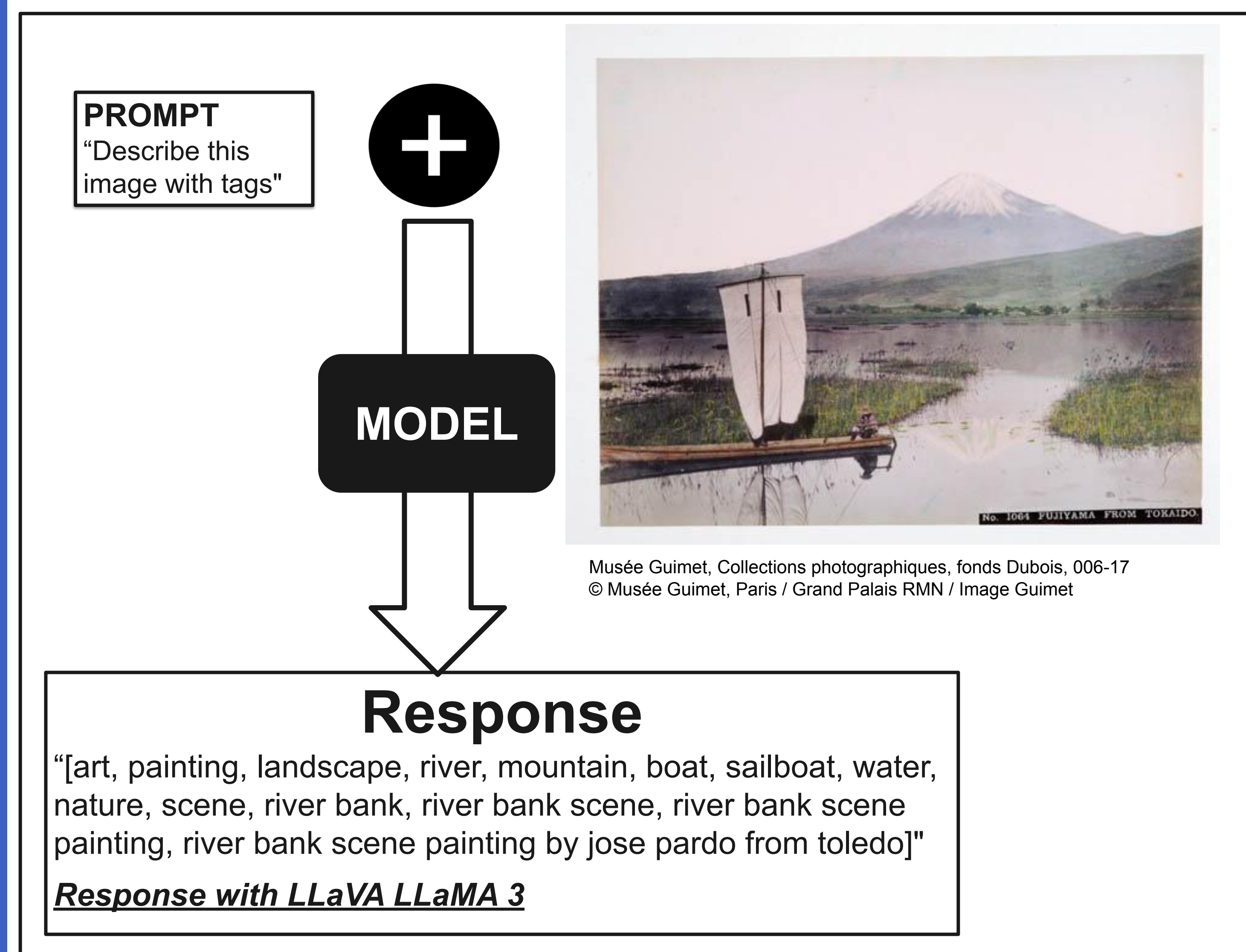
- Automatic analysis of photographs
- Creation of a freely accessible platform for exploring the corpus



QUESTIONS

- Are VLLM able to describe old photographs ?
- To what extent can the model's response be used by :
 - an IT specialist?
 - an archivist?
 - a historian?

EXPERIMENTS FOR IMAGE DESCRIPTION



EXPERIMENTS

- Promptfoo :
 - Prompt engineering and models comparison library
- Our contribution to promptfoo
 - Models implementation
 - Prompts definition
 - Prompt variables definition
 - Assertion tests
- Experimental conditions :
 - 275 annotated images
 - 190 different tags annotated
 - 4 Prompts:
 - Total : 8800 inferences

MODELS

- GPT-4o
- LLaVA
- BLIP
- CLIP
- Moondream
- Idefics2
- PaliGemma

PROMPTS

- Free description
- Free tags
- Free tags + formatted output
- Tags limited to a set provided by the user + formatted output

RESULTS

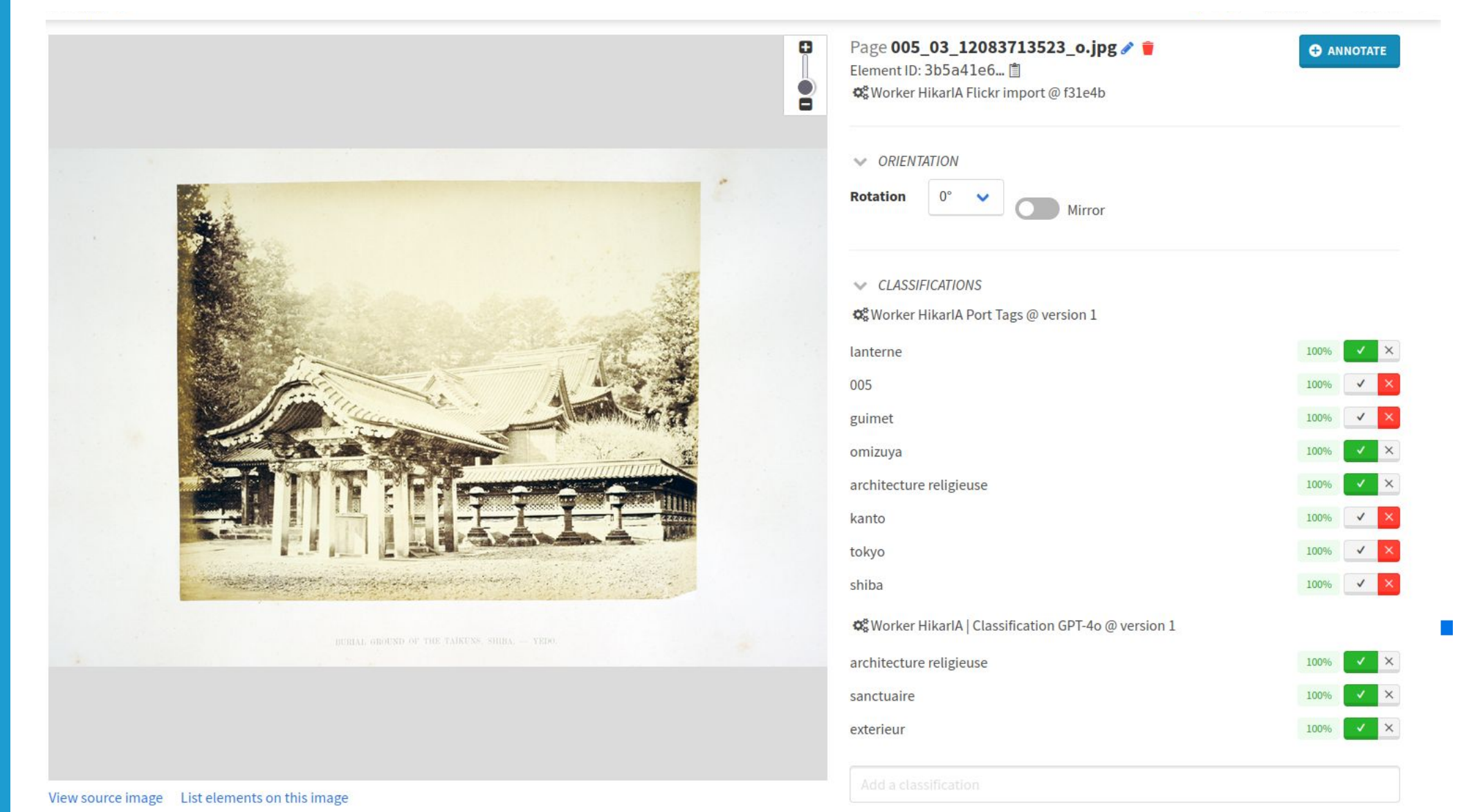
- GPT-4o results :
 - Best prompt : Tags limited to a set provided by the user and formatted output in French

Models	Precision	Recall	F1-score
GPT-4o Raw	52.1%	42.2%	44.9%
GPT-4o Human Validation	94.1%	53.8%	68.5%

- Other models obtain lower F1 scores (less than 1%)

CONCLUSION

RESULTS ON ARKINDEX



LIMITATIONS

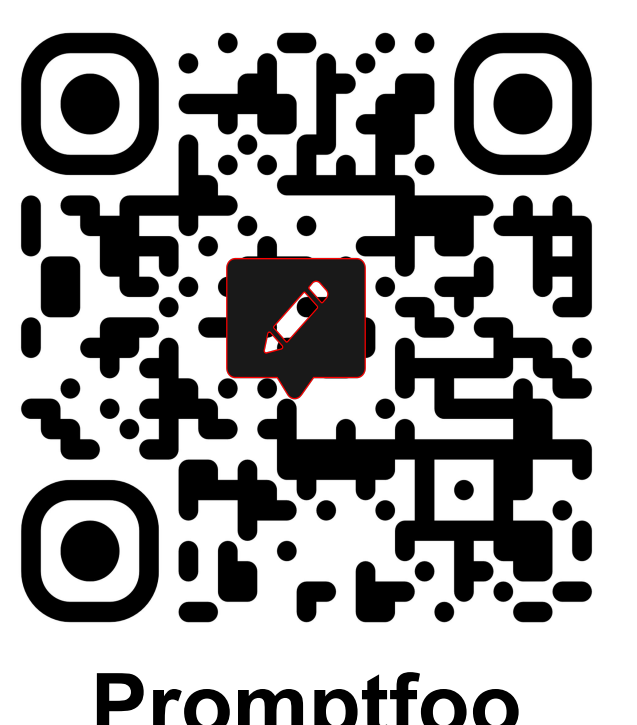
- Consistency in human annotation
- Tag-based evaluation is limited, as multiple words can carry the same semantic meaning
- Prompt specificity for each models
- No models as large as GPT-4o

POTENTIAL SOLUTION

- Get the information tag by tag
- Evaluate with Bert-score



Musée Guimet



Promptfoo

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